

Abderrahim BOUHENACHE

Born in Mila, Algeria (1995)

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PhD in Agronomy

Postdoc Researcher in soil-crop modelling at INRAE

for a more detailed *CURRICULUM VITAE*, click [here](#)



My skills lie in sustainable plant production and agroecology, with a strong background in field experimentation and agricultural surveys, and a particular interest in crop modeling. My expertise focuses on the design and assessment of cropping systems across Mediterranean (Algeria), temperate (France), and tropical (Zimbabwe) contexts. My research addresses issues of resource use efficiency, resilience, and adaptation to climate change.

PROFESSIONAL EXPERIENCE

French National Research Institute for Agriculture, Food and Environment (INRAE) - Mixed Research Unit Eco&Sols – Montpellier (France)

Apr. 2026 - Today
(Postdoc researcher contract)

Project: [ALAMOD](#)

Subject: Modelling of soil organic matter dynamics

- Evaluating STICS ability to simulate long-term C&N dynamics using LTEs datasets
- Evaluating and improving short- and long-term simulations of plant litter decomposition with STICS

The University of Reims Champagne-Ardenne (URCA) - Mixed Research Unit FARE – Reims (France) / Research Unit AIDA – Harare (Zimbabwe)

Sep. 2022 - Dec. 2025
(Temporary contract as PhD candidate)

Project: [RAIZ](#)

Subject: Impact of climate change on production and recycling of crop biomass and soil nitrogen use: a case study with maize under extreme rainfall events in sub-humid Zimbabwe

- Design and implementation of a field rainfall manipulation experiment in Zimbabwe
- Evaluation of crop performance drivers under contrasted management and rainfall
- Monitoring and analysis of *in-situ* surface crop residue decomposition dynamics
- Field assessment of nitrogen fertilizer fate in the soil–plant system via ¹⁵N tracing

L'Institut Polytechnique UniLaSalle - Research Unit AGHYLE - Beauvais (France)

Feb. 2022 - Aug. 2022
(6-month contract as studies engineer)

Project: [AD'METHA](#)

Subject: Agronomic, economic and environmental assessment of cropping systems designed to supply the methanization sector

- Data curation and analysis of the results of cropping system trials
- Participation in project activities (meetings, field days) and writing of technical reports

L'institut agro AGROCAMPUS OUEST - Mixed Research Unit SAS - Rennes (France)

Nov. 2020 - Nov. 2021
(12-month temporary contract)

Project: EVADIVER

Subject: Design and ex ante assessment of diversified cropping systems to increase the provision of ecosystem services of interest in a crop-livestock region.

- Bibliography and conceptual modeling of ecosystem services in diversified cropping systems
- Co-design of diversified cropping systems with local actors

French National Research Institute for Agriculture, Food and Environment (INRAE) - Mixed Research Unit AGIR - Toulouse (France)

Apr. 2020 - Sep. 2020
(6-month internship)

Project: ACCAF-OPERATE

Subject: Multi-criteria assessment of climate change adaptation scenarios for the sunflower crop

- Development of simple epidemiological models for sunflower diseases
- Simulation of risk trends according to adaptation strategies under future climate
- Multi-criteria assessment (production, environment and health risk) of strategies

EDUCATIONAL BACKGROUND

2026	University of Reims Champagne Ardenne - Reims (France) PhD in Agricultural Sciences
2020	CIHEAM-IAM / Paul-Valéry University Montpellier III - Montpellier (France) Master 2 « Agricultural management and Territories », ranking: 1/15 students
2019	Higher National Agronomic School (ENSA) - Algiers (Algeria) Master 2 « Genetic resources and improvement of plant production » ranking: 1/19 students Agronomy engineer « Plant production and improvement », ranking: 2/19 students

KNOWLEDGE AND SKILLS



Programming languages

- R ●●●●●
- C# ●●●●●
- GAMS ●●●●●



Crop modeling

- STICS & APSIM ●●●●● (in progress)
- CropSyst ●●●●●
- ModelBuilder (platform INRAE)



Statistics

- Statistica
- Tanagra
- R



Cartography

- QGIS
- ArcGIS
- R packages: sf, raster, whitebox



Languages

- English (professional)
- French (bilingual)
- Arabic (native language)



Soft skills

- Autonomous, serious and responsible
- Scientific rigor, team spirit
- Learning and synthesis skills

PUBLICATIONS

1. **Bouhenache, A.** Using rainfall exclusion systems to simulate crop responses to drought. *Nat Rev Earth Environ* (2026). <https://doi.org/10.1038/s43017-026-00784-0>
2. **Bouhenache A**, Lashermes G, Clivot H, Recous S, Chikowo R, Shumba A, Mazungunye H, Matimba E, Alavoine G, Delfosse O, Falconnier GN, Affholder F, Corbeels M, Cardinael R. Intra-seasonal rainfall patterns and extremes drive maize productivity and nitrogen use in sub-humid Zimbabwe. *Field Crops Res.* 2025; 334:110126. <https://doi.org/10.1016/j.fcr.2025.110126>
3. Kherif O, Seghouani M, Justes E, Plaza-Bonilla D, **Bouhenache A**, Zemmouri B, Dokukin P, Latati M. The first calibration and evaluation of the STICS soil-crop model on chickpea-based intercropping system under Mediterranean conditions. *Eur. J. Agron.* 2022; 133. <https://doi.org/10.1016/j.eja.2021.126449>
4. Kherif O, Seghouani M, Zemmouri B, **Bouhenache A**, Keskes MI, Yacer-Nazih R, Ouaret W, Latati M. Understanding the Response of Wheat-Chickpea Intercropping to Nitrogen Fertilization Using Agro-Ecological Competitive Indices under Contrasting Pedoclimatic Conditions. *Agronomy.* 2021; 11(6):1225. <https://doi.org/10.3390/agronomy11061225>

ADDITIONAL INFORMATION

- Representative of non-titular staff at the UPR AIDA and at the UMR SAS
- Master's grant, GAT program, CIHEAM-IAMM (2019)

CENTERS OF INTEREST

Football, travel and long-distance journeys, North African and Mediterranean cuisine.